Summary valuation methods
QF*

Starting point is a model for the 'basic assets' in the market, i.e. SDEs. In this summary we assume that these assets are a stock, with price process S, and a money-market account, with price process B. Given this model, this brief note summarizes the three methods to value a European option with payoff $C_T = h(S_T)$ at maturity T.

1 Partial differential equations

1. Restrict yourself to Markov trading strategies. So we can write
$$V_t = F(t, S_t). \tag{1}$$
2. Restrict yourself to self-financing trading strategies. So we have
$$\mathrm{d}V_t = \phi_t \,\mathrm{d}S_t + \psi_t \,\mathrm{d}B_t. \tag{2}$$
3. Derive representation for dynamics of V_t:
   - using (1) (and Itô's formula) we obtain $\mathrm{d}V_t = a_t \,\mathrm{d}t + b_t \,\mathrm{d}W_t$;
   - using (2) we obtain $\mathrm{d}V_t = c_t \,\mathrm{d}t + d_t \,\mathrm{d}W_t$.
4. Solving $a_t = c_t$ and $b_t = d_t$ yields a PDE for F.
5. To value the European option with payoff $C_T = h(S_T)$: solve the PDE from step 4 under the terminal condition $F(T, s) = h(s)$ for all $s > 0$. The fair price, at time t, is then given by $C_t = F(t, S_t)$.

2 Risk-neutral world

1. Pick a numéraire N (any of the basic assets with $N_t > 0$).
2. By the risk-neutral version of the First Fundamental Theorem of asset pricing (FFT) our basic market is free of arbitrage opportunities "if and only if" there exists a probability measure $\mathbb{Q}$ such that $\mathbb{Q} \sim \mathbb{P}$ and such that for all assets the discounted price, i.e. A_t/N_t, is a martingale under $\mathbb{Q}$.
3. Use the model to derive, for every basic asset, a SDE for the discounted price process, i.e.
$$\mathrm{d}\frac{A_t}{N_t} = a_t \,\mathrm{d}t + b_t \,\mathrm{d}W_t^{\mathbb{P}}.$$

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4. Introduce a new process $W^{\mathbb{Q}}$ by (we specify γ_t at a later stage)

$$\mathrm{d}W_t^{\mathbb{Q}} = \gamma_t \,\mathrm{d}t + \,\mathrm{d}W_t^{\mathbb{P}}, \qquad W_0^{\mathbb{Q}} = 0,$$

   and insert this into the previous display:

$$\mathrm{d}\frac{A_t}{N_t} = (a_t - b_t\gamma_t)\,\mathrm{d}t + b_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

   Choose $\gamma_t = a_t/b_t$, which yields:

$$\mathrm{d}\frac{A_t}{N_t} = b_t\,\mathrm{d}W_t^{\mathbb{Q}}.$$

   (For the Black-Scholes market with $N = B$ this gives $\gamma_t = (\mu - r)/\sigma$, the market price of risk.)

5. Apply Girsanov's theorem to find a probability measure $\mathbb{Q}$ that is equivalent to $\mathbb{P}$ and such that $W^{\mathbb{Q}}$ is a Brownian motion under $\mathbb{Q}$.

6. Conclude that all A_t/N_t are martingales under $\mathbb{Q}$.

7. To price the European option with payoff $C_T = h(S_T)$:

   - solving the SDEs (under $\mathbb{Q}$) typically yields $S_T = g(T, W_T^{\mathbb{Q}})$;
   - apply the FFT once more:

$$C_t = N_t\mathbb{E}_{\mathbb{Q}}\left[\frac{1}{N_T}\,h\Big(g(T,W_T^{\mathbb{Q}})\Big)\;\Big|\;\mathcal{F}_t\right], \qquad \text{in particular} \quad C_0 = N_0\mathbb{E}_{\mathbb{Q}}\left[\frac{1}{N_T}\,h\Big(g(T,W_T^{\mathbb{Q}})\Big)\right],$$

   and exploit that $W^{\mathbb{Q}}$ is a standard Brownian motion under $\mathbb{Q}$.

3 Pricing kernel

By the pricing kernel version of the FFT our market is free of arbitrage opportunities "if and only if" there exists some adapted process K_t such that $K_t > 0$ and such that for all basic assets in the market K_tA_t is a martingale (here A_t denotes the price of the asset).

1. Try $\mathrm{d}K_t = \mu_t^K\,\mathrm{d}t + \sigma_t^K\,\mathrm{d}W_t^{\mathbb{P}}$.

2. Use the model for the basic assets to solve for μ_t^K and σ_t^K, i.e. calculate $\mathrm{d}K_tA_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t^{\mathbb{P}}$ and ensure $a_t = 0$ for every basic asset.

3. To price the European option with payoff $C_T = h(S_T)$ apply the FFT once more:

$$C_t = \frac{1}{K_t}\mathbb{E}_{\mathbb{P}}[\,K_TC_T \mid \mathcal{F}_t\,], \qquad \text{in particular} \quad C_0 = \frac{1}{K_0}\mathbb{E}_{\mathbb{P}}[\,K_TC_T\,].$$